

Entropy rectifies the Brownian steps of kinesin

Yuichi Taniguchi^{1,2}, Masayoshi Nishiyama³, Yoshiharu Ishii¹ & Toshio Yanagida^{1,4,5}

Kinesin is a stepping motor that successively produces forward and backward 8-nm steps along microtubules. Under physiological conditions, the steps powering kinesin's motility are biased in one direction and drive various biological motile processes. The physical mechanism underlying the unidirectional bias of the kinesin steps is not fully understood. Here we explored the mechanical kinetics and thermodynamics of forward and backward kinesin steps by analyzing their temperature and load dependence. Results show that the frequency asymmetry between forward and backward steps is produced by entropy. Furthermore, the magnitude of the entropic asymmetry is $6 k_B T$, more than three times greater than expected from a current model, in which a mechanical conformational change within the kinesin molecular structure directly biases the kinesin steps forward. We propose that the stepping direction of kinesin is preferably caused by an entropy asymmetry resulting from the compatibility between the kinesin and microtubule interaction based on their polar structures.

Kinesin is an ATP-driven molecular motor^{1–4} that successively produces 8-nm steps along the microtubule lattice⁵. Almost all kinesin steps take place towards the plus end of a microtubule (that is, in the forward direction), with very few backward steps. The unidirectional occurrence of the steps is responsible for kinesin's directed motility, which is essential in various biological motile processes involving the kinesin motor, such as intracellular trafficking and cell division. A recent study has shown that processive backward stepping can be induced under very high backward loads (~ 15 pN)⁶, indicating that the stepping direction of kinesin is reversible. It is also known that both the forward and backward steps are coupled to ATP hydrolysis^{6–9}, which suggests that the step direction of kinesin is biased in one direction consuming the free energy of ATP hydrolysis.

A hand-over-hand mechanism has been proposed to describe kinesin steps and has been recently confirmed^{10–13}. In this mechanism, kinesin walks along the microtubule using the two head domains in an action similar to that of a human walking. The head that is detached from the microtubule undergoes a diffusive search for adjacent binding sites on the microtubule, and the direction in which the detached head lands in determines the direction of the stepping movement⁶. It is necessary to understand what biases the detached head to land in the forward direction so as to understand the mechanism underlying the unidirectional movement of kinesin. Structural studies of the kinesin molecule have led to the proposal that the detached head is biased toward the forward direction by a conformational change in the docking of the flexible, 3-nm-sized neck linker domain¹⁴ towards the partner head¹⁵. A recent spectroscopic study of the neck linker domain has shown that the free energy released by the docking is only $1\text{--}2 k_B T$ (ref. 16). However, some

groups argue that this energy is probably too low, because for kinesin to generate 8-nm steps against a load of 7 pN, up to $\sim 12 k_B T$ is required^{17,18}. In contrast, other groups argue that the small docking free energy would be enough to explain kinesin's unidirectional steps because the presence of the backward steps, especially under high load, decreases the required energy¹⁶. Thus, the mechanism underlying the unidirectional bias of kinesin steps remains a matter of active debate.

Studies on the experimental temperature and external load dependence of kinetics have been applied to various kinds of reactions to determine their physical mechanisms^{16,19,20}. Transition-state theory^{21–23} has been used to express the frequency of stochastic reaction events as a function of the activation free-energy barrier ($\Delta G^\ddagger$). The additional analysis of temperature (T) dependence of the reaction divides the activation free energy into an activation enthalpy term ($\Delta H^\ddagger$) and an activation entropy term ($\Delta S^\ddagger$), as described by the relationship $\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger$. The activation enthalpy can be interpreted as the total change in binding energy that is needed to activate the reaction²². The activation entropy, meanwhile, is interpreted as the change of all factors setting aside the change in binding energy necessary for activating the reaction. According to the Arrhenius rate law, this corresponds to a molecular steric factor²². The analysis of load (F) dependence offers an alternative way to analyze the free energy, where $\Delta G^\ddagger = Fd + \Delta G_{F=0}^\ddagger$ and d represents the magnitude of the mechanical transition during the reaction, which is called the characteristic distance¹⁹.

In this study, we have explored the mechanical kinetics and thermodynamics of kinesin forward and backward steps by analyzing their load and temperature dependence to determine the rectifying mechanism of the kinesin steps. The results show that the entropic

¹Soft Nanomachine Project, Japan Science and Technology Agency, 1-3, Yamadaoka, Suita, Osaka, 565-0871, Japan. ²Department of Mechanical Science and Bioengineering, Graduate School of Engineering Science, Osaka University, 1-3, Machikaneyama, Toyonaka, Osaka 560-8531, Japan. ³Department of Chemistry, Graduate School of Science, Kyoto University, Sakyo-ku, Kyoto, 606-8502, Japan. ⁴Laboratories for Nanobiology, Graduate School of Frontier Biosciences, Osaka University, 1-3, Yamadaoka and ⁵Department of Physiology and Biosignaling, Graduate School of Medicine, Osaka University, 2-2, Yamadaoka, Suita, Osaka, 565-0871, Japan. Correspondence should be addressed to T.Y. (yanagida@phys1.med.osaka-u.ac.jp).

Received 1 July; accepted 20 September; published online 9 October 2005; doi:10.1038/nchembio741

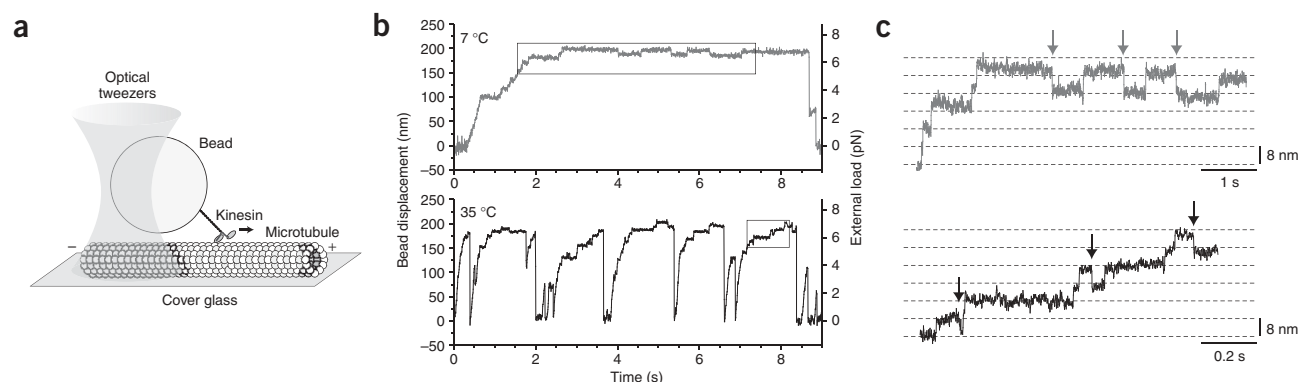


Figure 1 Nanometry of single kinesin molecules. (a) The optical trapping nanometry system (not to scale). The kinesin-coated bead captured by the optical tweezers was used as a probe for measuring the displacement of kinesin. (b) A typical displacement record by a single kinesin molecule at 7 °C (top, gray) and 35 °C (bottom, black) at saturating ATP concentrations ($= 1 \text{ mM}$). The external load, calculated using the trap stiffness (0.035 and 0.038 pN nm $^{-1}$), is indicated on the right. (c) Expansion of the displacement record. The boxed regions in b are expanded. The backward steps are indicated by arrows.

asymmetry is needed to determine the direction of kinesin steps. On the basis of these results, we propose a steric dependent model in which this entropic asymmetry rectifies the kinesin Brownian steps.

RESULTS

Measurement of 8-nm steps caused by single kinesin molecules

We used an optical trapping nanometry system^{24,25} to measure the individual stochastic stepping events caused by single kinesin molecules at various temperatures (Fig. 1a). In this system, single kinesin molecules were attached to polystyrene beads captured by an optical trap and were allowed to interact with the microtubule fixed on a glass surface. We traced the displacements of kinesin by monitoring the change in the position of a trapped bead with nanometer accuracy. When kinesin interacted with the microtubule, it moved at a constant speed against low load, and it gradually decreased its speed as the load increased until reaching a plateau before abruptly returning to the trap center (Fig. 1b). The force at the plateau was approximately $\sim 7 \text{ pN}$. At low temperature (7 °C), the speed decreased while the maximum force was unchanged, which is in good agreement with results from a previous study²⁶. The time course of kinesin displacement showed a distinctive stepwise pattern (Fig. 1c). The steps were primarily in the forward direction but occasionally in the backward direction (arrows in Fig. 1c). We measured the step size, defined as the difference between the average kinesin positions over a 5-ms interval before and after the step, at various temperatures (Supplementary Fig. 1). The distribution of the step size for the forward steps has a main peak at 8 nm but no peak at 16 nm ($< 0.2\%$ of all forward steps) for all temperatures, indicating that almost all 8-nm steps were resolved. For the backward steps, the distribution of the step size was concentrated at 8 nm just as for the forward steps, as shown in a previous work⁶. The step size obtained from the peak position in the histogram for forward and backward steps was not affected by temperature (Supplementary Fig. 1) or load⁹.

Directionality of steps and dwell-time distribution

The measured steps were grouped into three categories: the forward steps, the backward steps (step size $< 14 \text{ nm}$ = (the peak value of backward step histogram) + $(3 \times \text{s.d. of the peak distribution})$) and the detachments ($> 14 \text{ nm}$). The backward steps and detachments occurred more frequently at high loads. The fractions of forward steps, backward steps and detachments were plotted as a function of load at different temperatures (Fig. 2). The probability of forward steps

decreased from almost 1 at near-zero loads to 0.5 at about 8 pN, while both the backward steps and detachments increased. This feature was observed at all temperatures studied. The logarithm of the ratio of forward steps to backward steps decreased with load in a linear fashion (data not shown), as observed in previous studies^{6,9,27}. This result confirms that the stepping direction is reversible at various temperatures.

To examine the kinetics of the kinesin steps, we analyzed the distribution of the dwell time between steps. The dwell time was measured as the period from the point in time when the first step was half completed to the point in time when the next step was half completed. The distribution of the dwell times for the forward steps showed a single exponential decay with a rise phase in a short time range (Fig. 3a). The histogram could be well fitted to a two-rate-limiting transition scheme ($\chi^2(29) = 30$, $P > 0.1$, at 35 °C and 5.5–7.5 pN; for example, see Fig. 3a). The histogram, however, could not be fitted to a one-rate-limiting transition scheme ($\chi^2(29) = 110$, $P < 0.005$, for the same data set). We obtained similar results from the dwell-time histograms for all load and temperature conditions. We plotted the time constants obtained for the two rate-limiting transitions as a function of load (Fig. 3b). The time constant corresponding to the slow decay phase increased with load, whereas the time constant corresponding to the fast rise phase was constant over load, indicating that one of the two rate-limiting transitions is a load-dependent mechanical transition and the other is a load-independent

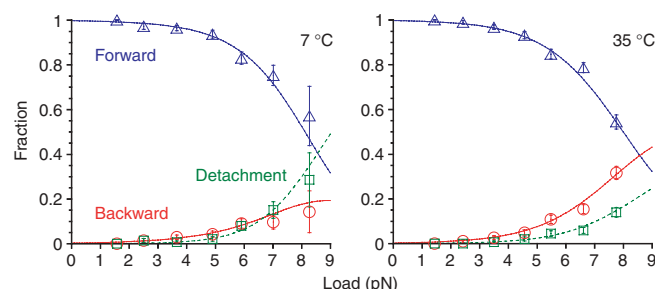


Figure 2 Load dependence of the proportion (mean $\pm$ s.e.m.) of forward steps (p_f , blue), backward steps (p_b , red) and detachments (p_d , green). The straight curves are calculated as $p_f = k_f/(k_f + k_b + k_d)$, $p_b = k_b/(k_f + k_b + k_d)$ and $p_d = k_d/(k_f + k_b + k_d)$ using the kinetic parameters summarized in Table 1.

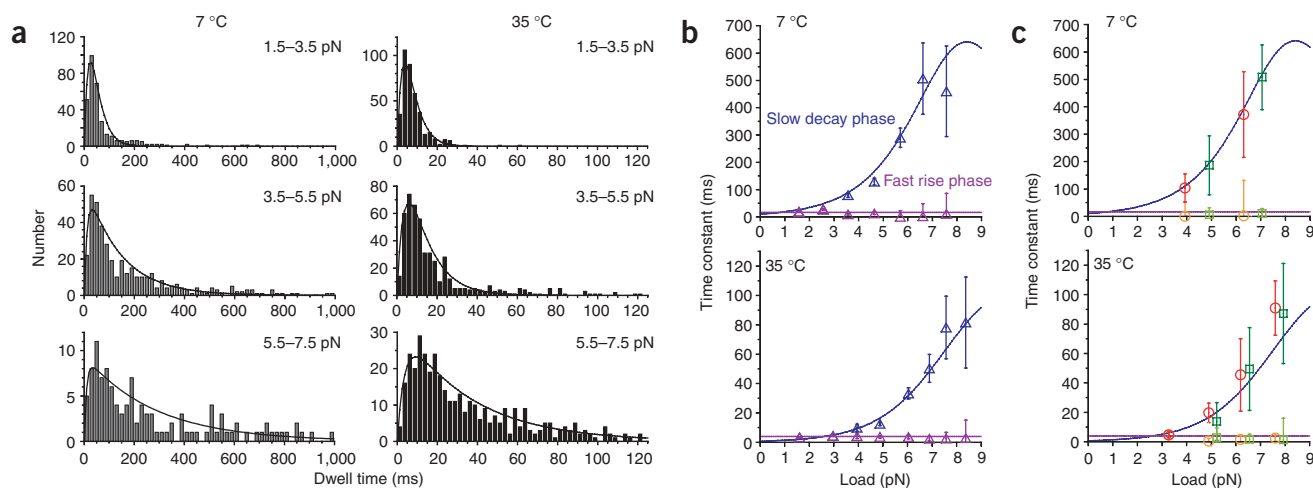


Figure 3 Dwell time at various loads and temperatures. **(a)** Histogram of dwell times of forward steps. **(b)** Load dependence of the time constants (mean $\pm$ s.e.m.) obtained by fitting the dwell-time distribution to a two-rate-limiting transition scheme. The time constants obtained at each temperature for the slow decay phase (blue) and the fast rise phase (purple) were plotted as a function of load. Solid curves were calculated as $1/(k_f + k_b + k_d)$ and $1/k_c$ using the kinetic parameters summarized in Table 1. **(c)** Time constants (mean $\pm$ s.e.m.) obtained for backward steps (red and orange) and detachments (green and yellow-green). The solid curves represent the lines of best fit of the forward steps in **b** and have been included for comparison.

biochemical transition^{9,17,19}. Both time constants increased as temperature decreased, in agreement with previous studies of microtubule gliding velocity and ATPase activity^{20,25,28,29}. The time constant for the slow phase doubled for every 10 °C reduction in temperature, whereas the time constant for the fast phase increased with a decrease in

temperature, but to a lesser extent. We also analyzed the histogram of dwell times for the backward steps and detachments in the same way (Fig. 3c, Supplementary Fig. 2). The time constants for the backward steps and detachments were essentially the same as those for the forward steps at each load and temperature, consistent with previous reports^{6,9}. This result suggests that the three events occur in parallel and start at the same state³⁰.

Mechanical kinetics of the kinesin step

On the basis of these results, we constructed a mechanical kinetics model for the step movement of kinesin (Fig. 4a). The kinesin steps occur in a sequence of load-independent biochemical transitions and a load-dependent mechanical transition. The stepping direction is determined at the mechanical transition, in which three pathways branch in parallel for forward steps, backward steps and detachments. The rate constants for these steps can be written as $k_f = k_m p_b$, $k_b = k_m p_b$ and $k_d = k_m p_d$, where k_m is the rate constant of the mechanical transition, an inverse of the time constant of the slow decay phase in the dwell-time histograms (Fig. 3). p_b , p_b and p_d are the fractions of the steps in the forward and backward direction and the detachments, respectively (Fig. 2). The rate constant of the biochemical transition, k_c , is obtained as the inverse of the time constants corresponding to the fast-rise phase of the dwell-time histogram (Fig. 3). The rate constants at the mechanical transition are plotted as a function of load in a semilogarithmic manner (Fig. 4b). The rate constant for the forward steps decreased linearly with load, but it did not change for the backward steps and increased only slightly with load for the detachments. The linear dependence of the plot was consistent with the Boltzmann-type dependence of the load as follows:

$$k_f(F, T) = k_f^0(T) \exp\left(-\frac{F d_f(T)}{k_B T}\right) \quad (1a)$$

$$k_b(F, T) = k_b^0(T) \exp\left(-\frac{F d_b(T)}{k_B T}\right) \quad (1b)$$

$$k_d(F, T) = k_d^0(T) \exp\left(-\frac{F d_d(T)}{k_B T}\right) \quad (1c)$$

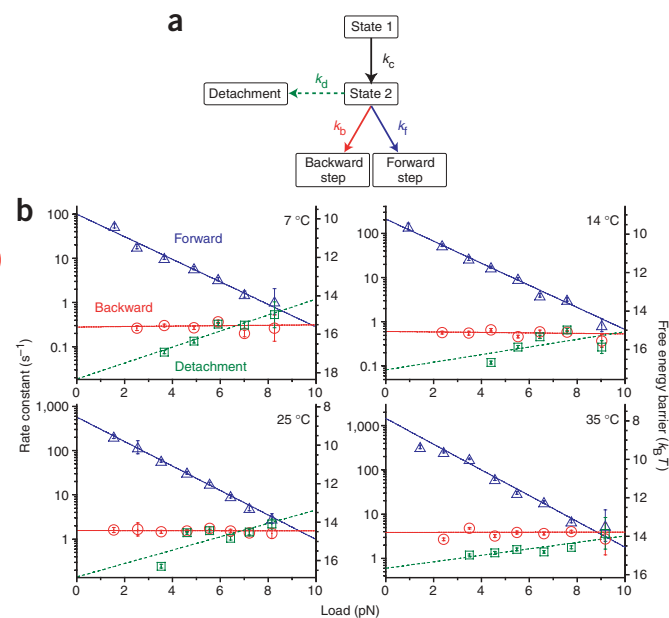


Figure 4 Kinetic analysis for kinesin steps. **(a)** Kinetic pathway of kinesin steps. Kinesin steps, either in the forward or backward direction, or detachments occur by passage through a sequence of load-independent 'biochemical' transitions (state 1 to state 2) and a load-dependent 'mechanical' transition (state 2 to forward step, backward step or detachments). **(b)** Load dependence of rate constants (mean $\pm$ s.e.m.) for the forward steps (blue), backward steps (red) and detachments (green). The lines are the lines of best fit to equations (1a), (1b) and (1c). All kinetic parameters determined are listed in Table 1.

Table 1 Parameters of mechanical kinetics of kinesin steps

Temperature T (K)	Rate constants				Characteristic distances		
	k_f^0 (s^{-1})	k_b^0 (s^{-1})	k_d^0 (s^{-1})	k_c (s^{-1})	d_f (nm)	d_b (nm)	d_d (nm)
280	100 ± 11	0.28 ± 0.04	0.019 ± 0.004	61 ± 4	2.4 ± 0.1	0.0 ± 0.1	-1.7 ± 0.2
287	209 ± 17	0.61 ± 0.06	0.082 ± 0.016	131 ± 15	2.3 ± 0.1	0.1 ± 0.1	-0.8 ± 0.1
298	544 ± 54	1.6 ± 0.2	0.14 ± 0.03	205 ± 11	2.6 ± 0.1	0.0 ± 0.1	-1.4 ± 0.1
308	$1,353 \pm 78$	3.8 ± 0.3	0.60 ± 0.07	257 ± 8	2.8 ± 0.1	0.0 ± 0.1	-0.7 ± 0.1

where F is the external load, k_B is the Boltzmann constant, T is the experimental temperature, $k_f^0(T)$, $k_b^0(T)$ and $k_d^0(T)$ are the rate constants at zero load, and $d_f(T)$, $d_b(T)$ and $d_d(T)$ are the respective characteristic distances, which can be interpreted as the distances over which the load acts during the forward step, backward step and detachment^{9,19}. Each rate constant increased with a rise in temperature; however, there was a greater effect on the mechanical transition rates than on the biochemical transition rate (Table 1). The characteristic distance of the forward steps is positive, whereas that of the backward steps is zero, indicating that the load-dependent reversibility of the stepping direction can be ascribed to just a decrease of the rate of forward steps with the load without any change in the rate of backward steps (Table 1).

Thermodynamic analysis of the kinesin step

The free-energy landscape has been determined by describing the stochastic kinesin steps in the forward and backward directions. According to transition-state theory^{21–23}, the stochastic stepping event is expressed as an activated free-energy barrier (Fig. 5a). The free-energy barriers for the forward ($\Delta G_f^\ddagger$) and backward steps ($\Delta G_b^\ddagger$) are described by the Boltzmann-type equations:

$$k_f(F, T) = AT \exp \left(-\frac{\Delta G_f^\ddagger(T) + Fd_f(T)}{k_B T} \right) \quad (2a)$$

$$k_b(F, T) = AT \exp \left(-\frac{\Delta G_b^\ddagger(T) + Fd_b(T)}{k_B T} \right) \quad (2b)$$

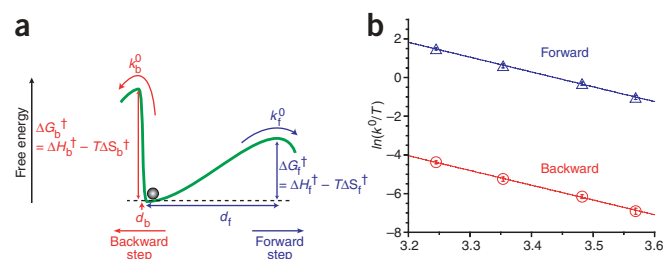


Figure 5 Thermodynamic analysis for kinesin steps (a) Free-energy landscape for kinesin steps. Kinesin undergoes thermal fluctuations around the equilibrium position and moves forward (to the right) or backward (to the left) when it acquires energy greater than the respective activation free-energy barriers, $\Delta G_f^\ddagger$ or $\Delta G_b^\ddagger$, as described by equations (2a) and (2b). The equilibrium position and energy barriers are separated by the corresponding characteristic distances, d_f and d_b . The free-energy barriers are composed of enthalpies, $\Delta H_f^\ddagger$ and $\Delta H_b^\ddagger$, and entropies, $\Delta S_f^\ddagger$ and $\Delta S_b^\ddagger$, related by the expressions $\Delta G_f^\ddagger = \Delta H_f^\ddagger - T\Delta S_f^\ddagger$ and $\Delta G_b^\ddagger = \Delta H_b^\ddagger - T\Delta S_b^\ddagger$. (b) Van't Hoff plots of the rate constants (mean $\pm$ s.e.m.) for the forward steps (blue) and backward steps (red). The dotted lines are the lines of best fit to equations (3a) and (3b). All thermodynamic parameters determined are listed in Table 2.

where the pre-exponential factor, AT , is assumed to equal the diffusion-limited rate²³ (see Methods). From the rate constants at zero load, k_f^0 and k_b^0 , we calculated the free-energy barriers for the forward and backward steps as $\Delta G_f^\ddagger = 8.7 \pm 0.1 k_B T$ and $\Delta G_b^\ddagger = 14.7 \pm 0.1 k_B T$ at $T = 298$ K, respectively (Table 2). The results indicate that the directional preference of kinesin steps can be ascribed to a free-energy difference of $6 k_B T$ ($= \Delta G_b^\ddagger - \Delta G_f^\ddagger$). Notably, the free energy required for kinesin's directional preference is more than three times greater than the free energy released by the docking in the neck linker domain ($1-2 k_B T$; ref. 16). This indicates that the unidirectional bias of kinesin is caused, or at least assisted, by an additional mechanism.

The free-energy barriers can be divided into an activation enthalpy ($\Delta H_f^\ddagger$ and $\Delta H_b^\ddagger$) and activation entropy ($\Delta S_f^\ddagger$ and $\Delta S_b^\ddagger$), where $\Delta G_f^\ddagger = \Delta H_f^\ddagger - T\Delta S_f^\ddagger$ and $\Delta G_b^\ddagger = \Delta H_b^\ddagger - T\Delta S_b^\ddagger$, respectively. We determined the enthalpic and entropic contributions from the temperature dependence of the rate constants of the forward and backward steps at zero load (Fig. 5b). The result shows that the enthalpy barrier for the movements, which is represented by the slope of the Van't Hoff plots, was the same for both the forward and backward direction ($\Delta H_f^\ddagger = \Delta H_b^\ddagger = 18 \pm 1 k_B T$ ($T = 298$ K)) (Table 2). The entropy barrier was greater for the forward step ($\Delta S_f^\ddagger = 9.9 \pm 1.1 k_B$) than for the backward step ($\Delta S_b^\ddagger = 3.9 \pm 1.4 k_B$). Thus, the free-energy difference originates not from the difference in enthalpy but solely from the differences in entropy, indicating that kinesin steps are directed by entropy alone. This suggests that a steric geometrical molecular effect around kinesin resulting in the entropy asymmetry²² is responsible for determining the stepping direction of kinesin.

DISCUSSION

In this study we explored the mechanical kinetics and thermodynamics of the kinesin steps. Our main conclusion from the thermodynamic analysis is that the stepping direction is determined solely by an entropic or steric²¹ asymmetry of $6 k_B T$. This free-energy difference cannot be explained solely by the free energy from the docking of the neck linker ($1-2 k_B T$; ref. 16). Furthermore, the mechanical kinetic analysis showed asymmetric characteristic distances for the forward and backward steps ($d_f = 2-3$ nm and $d_b = 0$ nm). This suggests that an asymmetric mechanical element contributes to determining in which direction kinesin will step. We propose a plausible model that depends on the neck linker model

Table 2 Thermodynamic parameters of kinesin steps

	Free energy ($\Delta G^\ddagger$) ($k_B T_0$)	Enthalpy ($\Delta H^\ddagger$) ($k_B T_0$)	Entropy ($\Delta S^\ddagger$) (k_B)
Forward step	8.7 ± 0.1	18.3 ± 1.1	9.9 ± 1.1
Backward step	14.7 ± 0.1	18.2 ± 1.4	3.9 ± 1.4
Energy difference	-6.0	0.1	6.0

$T_0 = 298$ K.

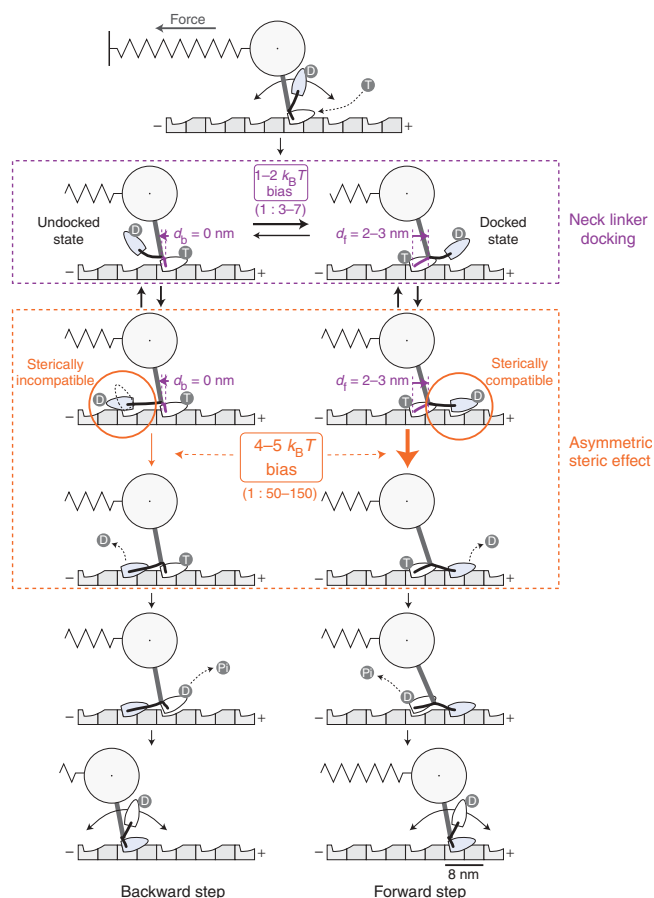


Figure 6 A model for the rectifying mechanism of stepping direction. Initially, one head with ADP bound (gray) detaches from the microtubule and diffuses on the basis of the partner head with no nucleotide bound (white). After ATP binds to the partner head, an equilibrium between the undocked and docked state of neck linker domain emerges. The equilibrium is biased to the docked state by $1-2 k_B T$, which yields a smaller directional bias of the detached head landing in the forward direction (purple box). In the undocked state, the backward landing of the detached head is geometrically unfavorable, because the detached head and the microtubule binding site are sterically incompatible owing to their different orientation. This steric, or entropic, effect adds another $4-5 k_B T$ to the directional bias of the head landing (orange box). After the head has landed, the alternative head takes off from the microtubule to complete steps, returning to the initial stage of the cycle. The external load pulls the branch point at the ends of neck linkers backward. The forward steps are caused by a lateral diffusion of the branch point in the forward direction by $d_f = 2-3 \text{ nm}$. In contrast, the backward steps are caused by a rotational diffusive search of the detached head for the reverse direction; therefore, the lateral diffusion is not necessary, resulting in $d_b = 0 \text{ nm}$. Letters in the circle at the head represent the prospective binding nucleotide: T, ATP; D, ADP; P, phosphate.

but incorporates an asymmetric steric component that explains the energy difference and the mechanical properties of process (Fig. 6).

In a hand-over-hand model, as previously described⁶, the forward and backward steps begin from a common intermediate, in which the attached head lacks a nucleotide, while the partner detached head binds ADP¹⁰. ATP binding to the attached head causes the conformational docking of the neck linker domain¹⁵ resulting in a biasing of the diffusive movement of the detached head towards the microtubule plus end^{31,32}. However, this bias is small, only $1-2 k_B T$, as mentioned above. The additional $4-5 k_B T$ is most likely to originate from a sterically restricted geometry between the kinesin head and the microtubule binding site that contributes to the entropic asymmetry. Owing to the short length of the neck linker domain, the orientation distribution of the detached head relative to the orientation of the microtubule binding site should differ when it is located in the forward or backward position³³. We argue that, in the forward position, the most favorable orientation of the detached head should coincide with the orientation of the binding site on the microtubules. In contrast, in the backward position the most common head orientation is incompatible with the binding site, and thus the probability of attachment is lowered. The attachment of the detached head causes the release of ADP from the head to be accelerated by the generation of an internal strain between the heads^{6,34}. This subsequently leads to the detachment of the alternative head, which occurs to complete either the forward or backward step¹¹.

The load-dependent modulation of the stepping direction can also be explained within the hand-over-hand framework (Fig. 6). We hypothesize that the characteristic distances we determined represent the lateral diffusive displacement of the coiled-coil domain between

two heads that is necessary to produce the forward and backward landing of the head, reflecting a scheme that is essentially the same as a previous model¹⁷. One interpretation is that when forward landing occurs, the coiled coil diffuses forward by $2-3 \text{ nm}$ ($= d_f$) against the load. This is most likely to correspond to the docked state of the neck linker domain^{15,16}. Because $d_b = 0 \text{ nm}$, it is likely that the backward landing can occur only if activated by factors other than the lateral diffusion of the coiled-coil domain, such as a rotational diffusive search of the detached head for the reverse direction. Thus, the external load will act on the forward steps but not the backward steps. **Detachment from the microtubule would be activated by the backward diffusion of the coiled-coil domain ($d_d = -1 \text{ nm}$), presumably as previously seen³⁵.** This hypothesis is further supported by the fact that the characteristic distance of the forward step, $2-3 \text{ nm}$, is similar to the length of the neck linker domain^{14,16}. The entropic rectification mechanism of the stepping direction may be a common feature of various kinds of motor proteins such as myosin^{36,37} and dynein.

METHODS

Sample preparation. Kinesin and tubulin were obtained from bovine brains as described²⁴. Tubulin was labeled with tetramethylrhodamine succinimidyl ester and then polymerized into microtubules. Assays were carried out in a solution containing 2 mM MgCl_2 , 1 mM EGTA , $10 \mu\text{M taxol}$, 0.2 mg mL^{-1} casein, 1 mM ATP , 80 mM PIPES (pH 6.8) and antiphotobleaching reagents (10 mM glucose , $9 \mu\text{g mL}^{-1}$ catalase, $50 \mu\text{g mL}^{-1}$ glucose oxidase and 0.5% 2-mercaptoethanol). Carboxylate-modified polystyrene beads ($0.22 \mu\text{m}$ in diameter) were precoated with 10 mg mL^{-1} casein and mixed with kinesin at a kinesin/bead molar ratio of 1:1. To confirm that the displacements of the beads were caused by single kinesin molecules, the probability of beads that bind and move processively on the microtubule as a function of kinesin/bead molar ratio were examined. The experiments described here used a kinesin/bead ratio low enough that less than 33% of the beads were moving. Assuming Poisson statistics, >95% of the moving beads were estimated to be driven by a single kinesin molecule³⁸. The assay was performed at various room temperatures ($7, 14, 25$ and 35°C), which were regulated by an air conditioner with an accuracy of 1°C .

Nanometry. The displacement of the beads was measured using a quadrant photodiode (S4349; Hamamatsu Photonics) and recorded at a sampling rate of 40 kHz with a bandwidth of 20 kHz . The raw traces of the bead were passed through a low-pass filter with a bandwidth of 1 kHz , which allowed the discrimination of individual 8-nm steps when the force was greater than 1 pN . The step size is defined as the difference between the average kinesin positions

over a 5-ms interval before and after the step. We confirmed that there was no other significant step (>1 nm) in the dwell time between the 8-nm steps at a temporal resolution of 1 ms, by averaging the traces after synchronizing the rising phases of 8-nm steps at the start and end of the dwell time and verifying its slope⁹. The dwell time was measured as the period from the point in time when the first step was half completed to the point in time when the next step was half completed. It is unlikely that the rise phase at the fast time range in the dwell-time histogram (Fig. 3a) was the result of the fast events being missed as a result of the insufficient time resolution of our measurement system, because 16-nm steps were rarely observed in the step-size histogram (see **Supplementary Fig. 1**). The distribution of dwell times assuming a one- or two-rate-limiting transition scheme for steps is given by $nb\tau_1^{-1}\exp(-\tau/\tau_1)$ or $nb(t_1t_2(1/t_2 - 1/t_1))^{-1}(\exp(-\tau/t_1) - \exp(-\tau/t_2))$, where τ is the dwell time, n is the number of steps, b is the bin width and t_1 and t_2 are the time constants. All parameter errors were estimated according to the law of propagation of uncertainty and the method of constant χ^2 boundaries as confidence limits^{39–41}. The numbers of total steps analyzed were 1,011, 1,497, 1,324 and 1,693 at 7, 14, 25 and 35 °C, respectively.

Free-energy landscape. The free-energy barriers for steps were calculated using equations (2a) and (2b). The pre-exponential factor, AT , as expressed in these equations, was assumed to be the diffusion-limiting rate²³ estimated from $k_B T / (3\pi\eta r x_0^2)$, the reciprocal of the Kramers first-passage time⁴² for a kinesin molecule to diffuse $x_0 = 8$ nm, where η is the viscosity of water depending on temperature and r is the radius of a kinesin molecule, which we estimated to be 2.6 nm assuming the kinesin was a sphere with a volume equal to that of the crystal structure of the kinesin motor domain ($7 \times 4.5 \times 4.5$ nm)⁴³. From this estimation, AT was determined to be 1.7, 2.1, 2.9 and 3.8 μs^{-1} at 7, 14, 25 and 35 °C, respectively. These values are consistent with the kinesin free-diffusion rate observed using a confocal microscope⁴⁴. The enthalpy and entropy barriers for the forward and backward steps were determined by fitting the Van't Hoff plots of rate constants (Fig. 5b) to equations:

$$\ln(k_f^0(T)/T) = \ln(A) + \frac{\Delta S_f^\ddagger}{k_B} - \frac{\Delta H_f^\ddagger}{k_B T} \quad (3a)$$

$$\ln(k_b^0(T)/T) = \ln(A) + \frac{\Delta S_b^\ddagger}{k_B} - \frac{\Delta H_b^\ddagger}{k_B T} \quad (3b)$$

Note: Supplementary information is available on the Nature Chemical Biology website.

ACKNOWLEDGMENTS

We thank F. Oosawa, T. Kodama, K. Sekimoto, H. Higuchi, A. Ishijima, H. Kojima, T. Ariga and colleagues at Osaka University and Japan Science and Technology Agency for valuable discussions and P. Karagiannis and J. West for carefully revising the manuscript.

COMPETING INTERESTS STATEMENT

The authors declare that they have no competing financial interests.

Published online at <http://www.nature.com/naturechemicalbiology/>
Reprints and permissions information is available online at <http://npg.nature.com/reprintsandpermissions/>

- Vale, R.D. & Milligan, R.A. The way things move: looking under the hood of molecular motor proteins. *Science* **288**, 88–95 (2000).
- Howard, J. Molecular motors: structural adaptations to cellular functions. *Nature* **389**, 561–567 (1997).
- Cross, R.A. The kinetic mechanism of kinesin. *Trends Biochem. Sci.* **29**, 301–309 (2004).
- Endow, S. Kinesin motors as molecular machines. *Bioessays* **25**, 1212–1219 (2003).
- Svoboda, K., Schmidt, C.F., Schnapp, B.J. & Block, S.M. Direct observation of kinesin stepping by optical trapping interferometry. *Nature* **365**, 721–727 (1993).
- Carter, N.J. & Cross, R.A. Mechanics of the kinesin step. *Nature* **435**, 308–312 (2005).
- Hua, W., Young, E.C., Fleming, M.L. & Gelles, J. Coupling of kinesin steps to ATP hydrolysis. *Nature* **388**, 390–393 (1997).

- Schnitzer, M.J. & Block, S.M. Kinesin hydrolyses one ATP per 8-nm step. *Nature* **388**, 386–390 (1997).
- Nishiyama, M., Higuchi, H. & Yanagida, T. Chemomechanical coupling of the forward and backward steps of single kinesin molecules. *Nat. Cell Biol.* **4**, 790–797 (2002).
- Hackney, D.D. Evidence for alternating head catalysis by kinesin during microtubule-stimulated ATP hydrolysis. *Proc. Natl. Acad. Sci. USA* **91**, 6865–6869 (1994).
- Kaseda, K., Higuchi, H. & Hirose, K. Alternate fast and slow stepping of a hetero-dimeric kinesin molecule. *Nat. Cell Biol.* **5**, 1079–1082 (2003).
- Yildiz, A., Tomishige, M., Vale, R.D. & Selvin, P.R. Kinesin walks hand-over-hand. *Science* **303**, 676–678 (2004).
- Asbury, C.L., Fehr, A.N. & Block, S.M. Kinesin moves by an asymmetric hand-over-hand mechanism. *Science* **302**, 2130–2134 (2003).
- Case, R.B., Pierce, D.W., Hom-Booher, N., Hart, C.L. & Vale, R.D. The directional preference of kinesin motors is specified by an element outside of the motor catalytic domain. *Cell* **90**, 959–966 (1997).
- Rice, S. *et al.* A structural change in the kinesin motor protein that drives motility. *Nature* **402**, 778–784 (1999).
- Rice, S. *et al.* Thermodynamic properties of the kinesin neck-region docking to the catalytic core. *Biophys. J.* **84**, 1844–1854 (2003).
- Schnitzer, M.J., Visscher, K. & Block, S.M. Force production by single kinesin motors. *Nat. Cell Biol.* **2**, 718–723 (2000).
- Block, S.M., Asbury, C.L., Shaevitz, J.W. & Lang, M.J. Probing the kinesin reaction cycle with a 2D optical force clamp. *Proc. Natl. Acad. Sci. USA* **100**, 2351–2356 (2003).
- Wang, M. *et al.* Force and velocity measured for single molecules of RNA polymerase. *Science* **282**, 902–907 (1998).
- Mazumdar, M. & Cross, R.A. Engineering a lever into the kinesin neck. *J. Biol. Chem.* **273**, 29352–29359 (1998).
- Eyring, H. The activated complex in chemical reactions. *J. Chem. Phys.* **3**, 107–115 (1935).
- Atkins, P.W. *Physical Chemistry* 3rd edn. Ch. 30, 737–761 (Oxford University Press, Oxford, UK, 1987).
- Yang, W.Y. & Gruebele, M. Folding at the speed limit. *Nature* **423**, 193–197 (2003).
- Kojima, H., Muto, E., Higuchi, H. & Yanagida, T. Mechanics of single kinesin molecules measured by optical trapping nanometry. *Biophys. J.* **73**, 2012–2022 (1997).
- Nishiyama, M. *et al.* Substeps within the 8-nm step of the ATPase cycle of single kinesin molecules. *Nat. Cell Biol.* **3**, 425–428 (2001).
- Kawaguchi, K. & Ishiwata, S. Temperature dependence of force, velocity, and processivity of single kinesin molecules. *Biochem. Biophys. Res. Commun.* **272**, 895–899 (2000).
- Nishiyama, M., Higuchi, H., Ishii, Y., Taniguchi, Y. & Yanagida, T. Single molecule processes on the stepwise movement of ATP-driven molecular motors. *Biosystems* **71**, 145–156 (2003).
- Böhm, K. *et al.* Effect of temperature on kinesin-driven microtubule gliding and kinesin ATPase activity. *FEBS Lett.* **466**, 59–62 (2000).
- Kawaguchi, K. & Ishiwata, S. Thermal activation of single kinesin molecules with temperature pulse microscopy. *Cell Motil. Cytoskeleton* **49**, 41–47 (2001).
- Fersht, A.R. *Structure and Mechanism in Protein Science* (W.H. Freeman, New York, 1999).
- Hirose, K., Lockhart, A., Cross, R.A. & Amos, L.A. Three-dimensional cytoelectron microscopy of dimeric kinesin and ncd motor domains on microtubules. *Proc. Natl. Acad. Sci. USA* **93**, 9539–9544 (1996).
- Hirose, K., Löwe, J., Alonso, M., Cross, R.A. & Amos, L.A. Congruent docking of dimeric kinesin and ncd into three-dimensional electron cryomicroscopy maps of microtubule-motor ADP complexes. *Mol. Biol. Cell* **10**, 2063–2074 (1999).
- Mandelkow, E. & Hoenger, A. Structures of kinesin and kinesin-microtubule interactions. *Curr. Opin. Cell Biol.* **11**, 34–44 (1999).
- Uemura, S. & Ishiwata, S. Loading direction regulates the affinity of ADP for kinesin. *Nat. Struct. Biol.* **10**, 308–311 (2003).
- Kawaguchi, K., Uemura, S. & Ishiwata, S. Equilibrium and transition between single- and double-headed binding of kinesin as revealed by single-molecule mechanics. *Biophys. J.* **84**, 1103–1113 (2003).
- Ali, M.Y. *et al.* Unconstrained steps of myosin VI appear longest among known molecular motors. *Biophys. J.* **86**, 3804–3810 (2004).
- Kitamura, K., Tokunaga, M., Esaki, S., Iwane, A.H. & Yanagida, T. Mechanism of muscle contraction based on stochastic properties of single actomyosin motors observed *in vitro*. *Biophysics* **1**, 1–19 (2005).
- Svoboda, K. & Block, S.M. Force and velocity measured for single kinesin molecules. *Cell* **77**, 773–784 (1994).
- William, H.P., Saul, A.T., William, T.V. & Brian, P.F. *Numerical Recipes in C* 2nd edn. (Cambridge University Press, New York, 1992).
- Rief, M. *et al.* Myosin-V stepping kinetics: a molecular model for processivity. *Proc. Natl. Acad. Sci. USA* **97**, 9482–9486 (2000).
- Altman, D., Sweeney, H.L. & Spudis, J.A. The mechanism of myosin VI translocation and its load-induced anchoring. *Cell* **116**, 737–749 (2004).
- Howard, J. *Mechanics of Motor Proteins and the Cytoskeleton* (Sinauer Associates, Sunderland, Massachusetts, 2001).
- Kull, F.J. *et al.* Crystal structure of the kinesin motor domain reveals a structural similarity to myosin. *Nature* **380**, 550–555 (1996).
- Tadakuma, H., Yamaguchi, J., Ishihara, Y. & Funatsu, T. Imaging of single fluorescent molecules using video-rate confocal microscopy. *Biochem. Biophys. Res. Commun.* **287**, 323–327 (2001).